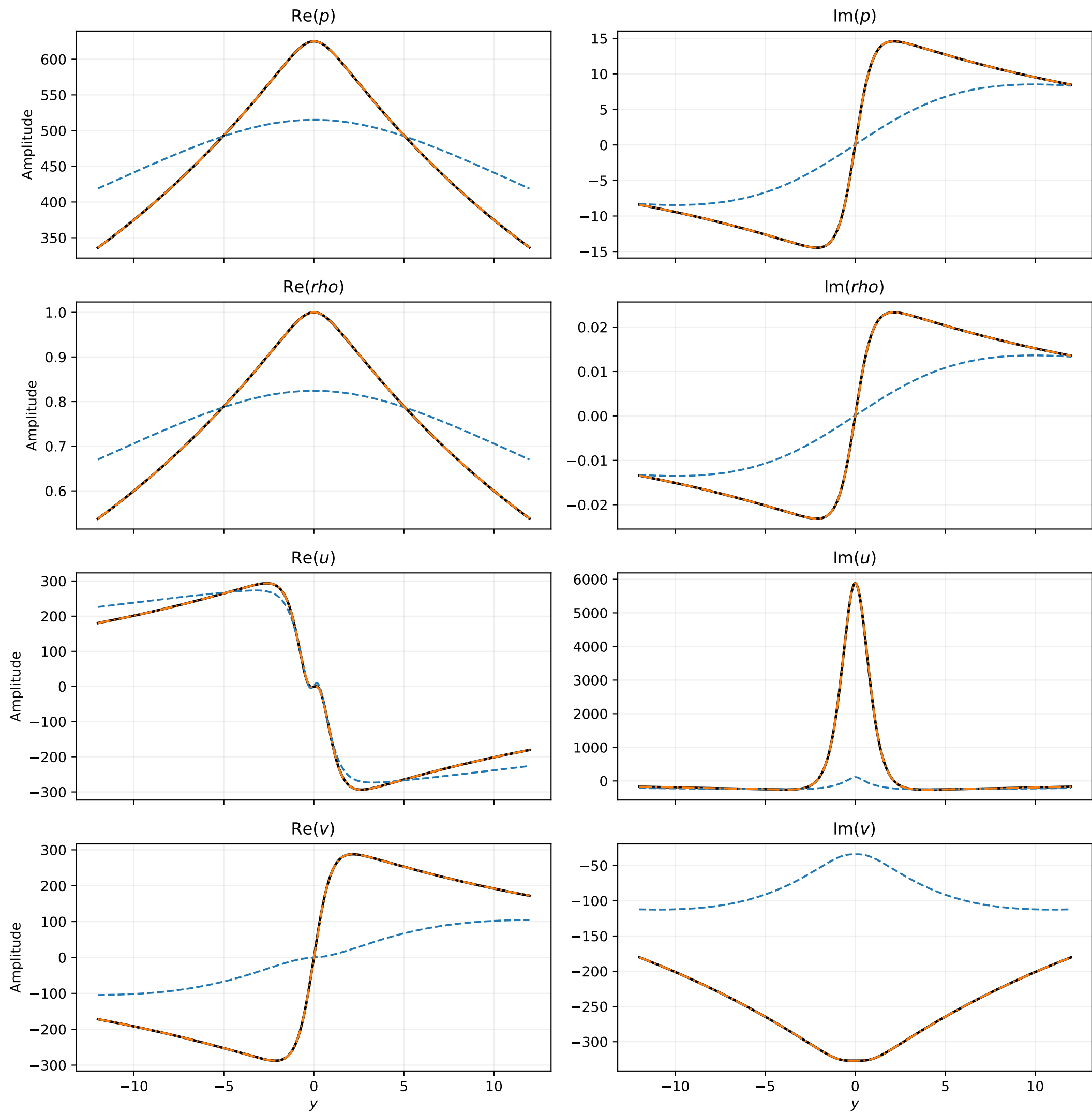


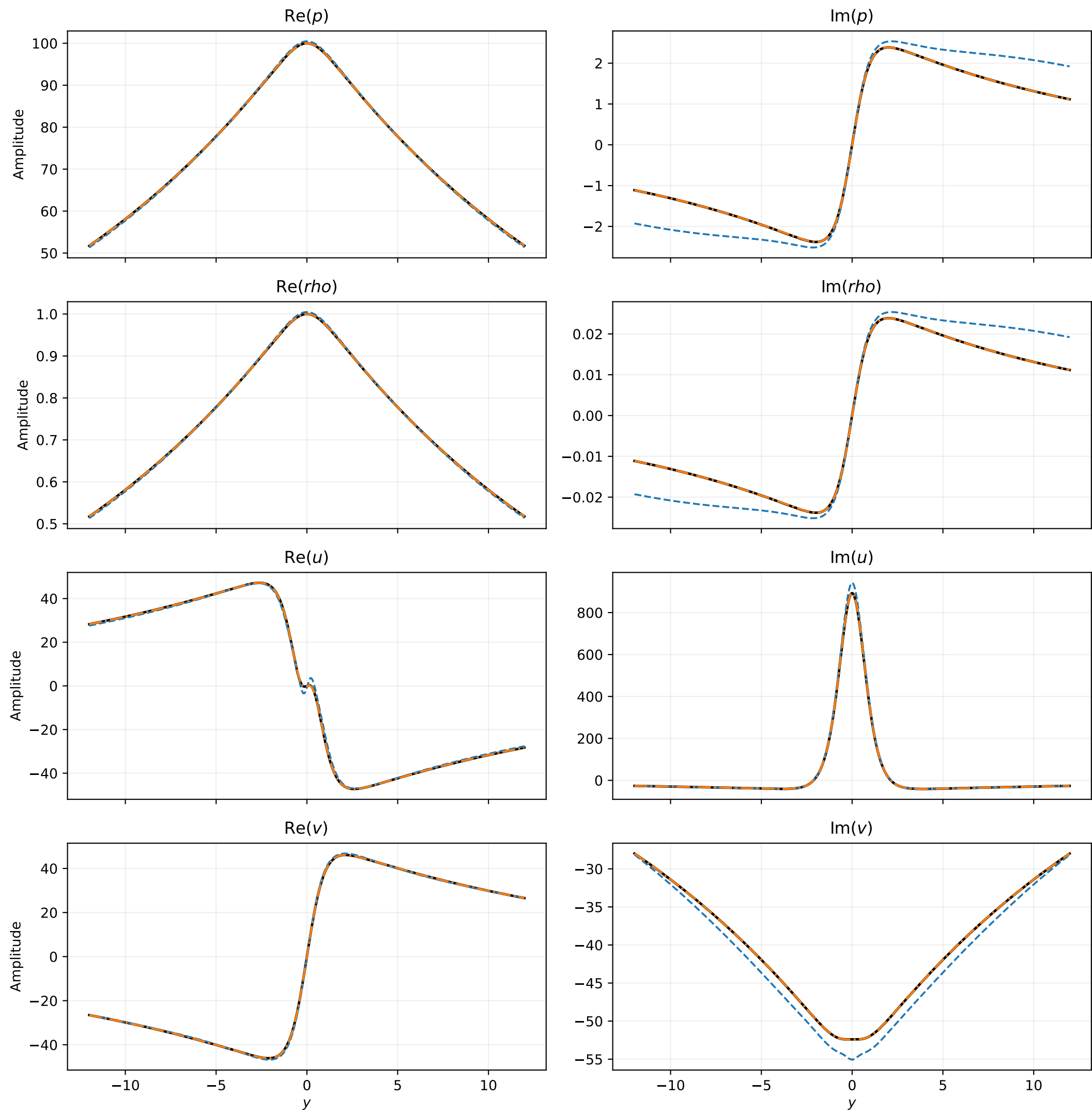
$$M = 0.04000, \eta = 0.05500, \alpha = 0.054956, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



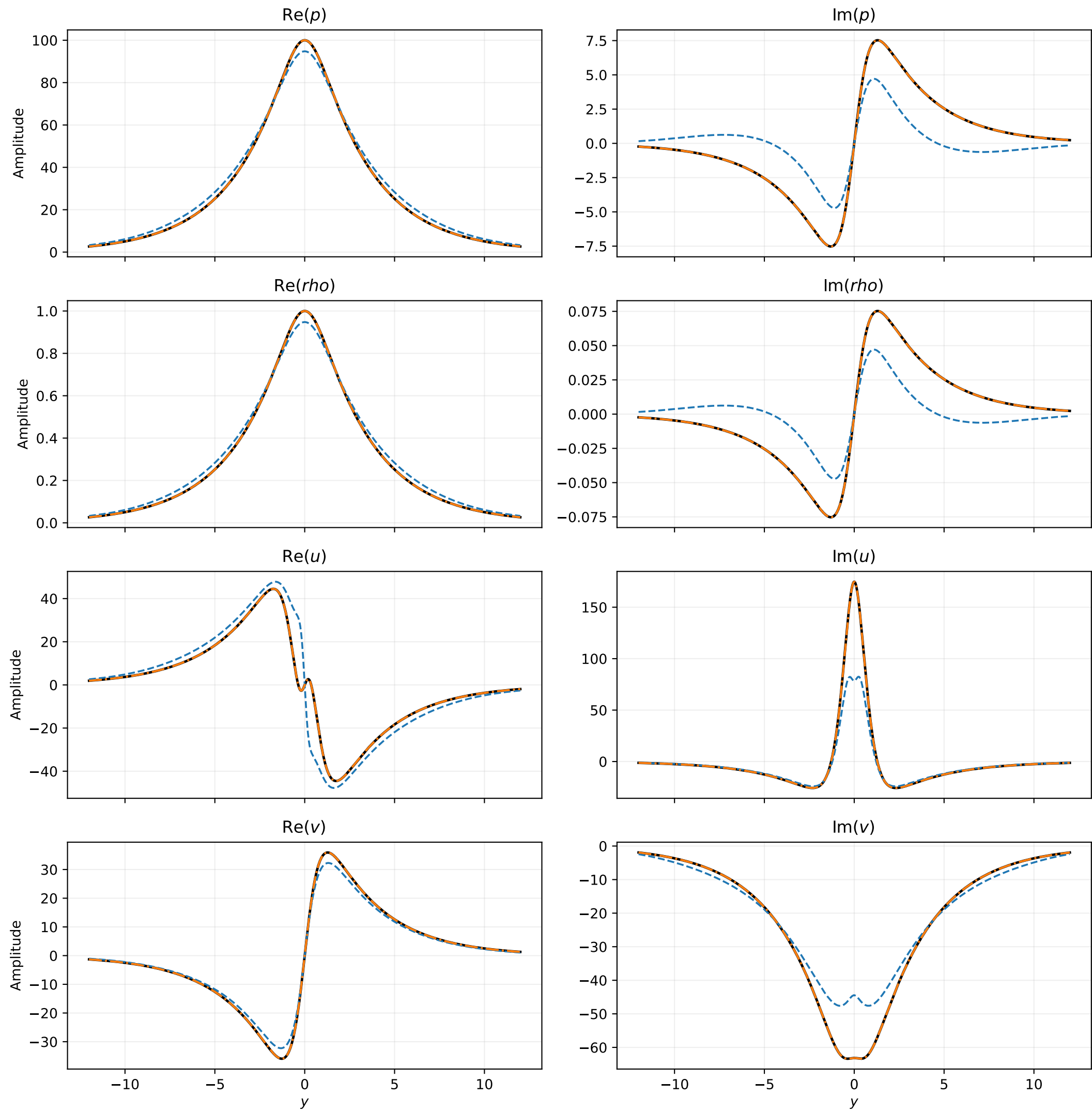
$$M = 0.10000, \eta = 0.05875, \alpha = 0.058456, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



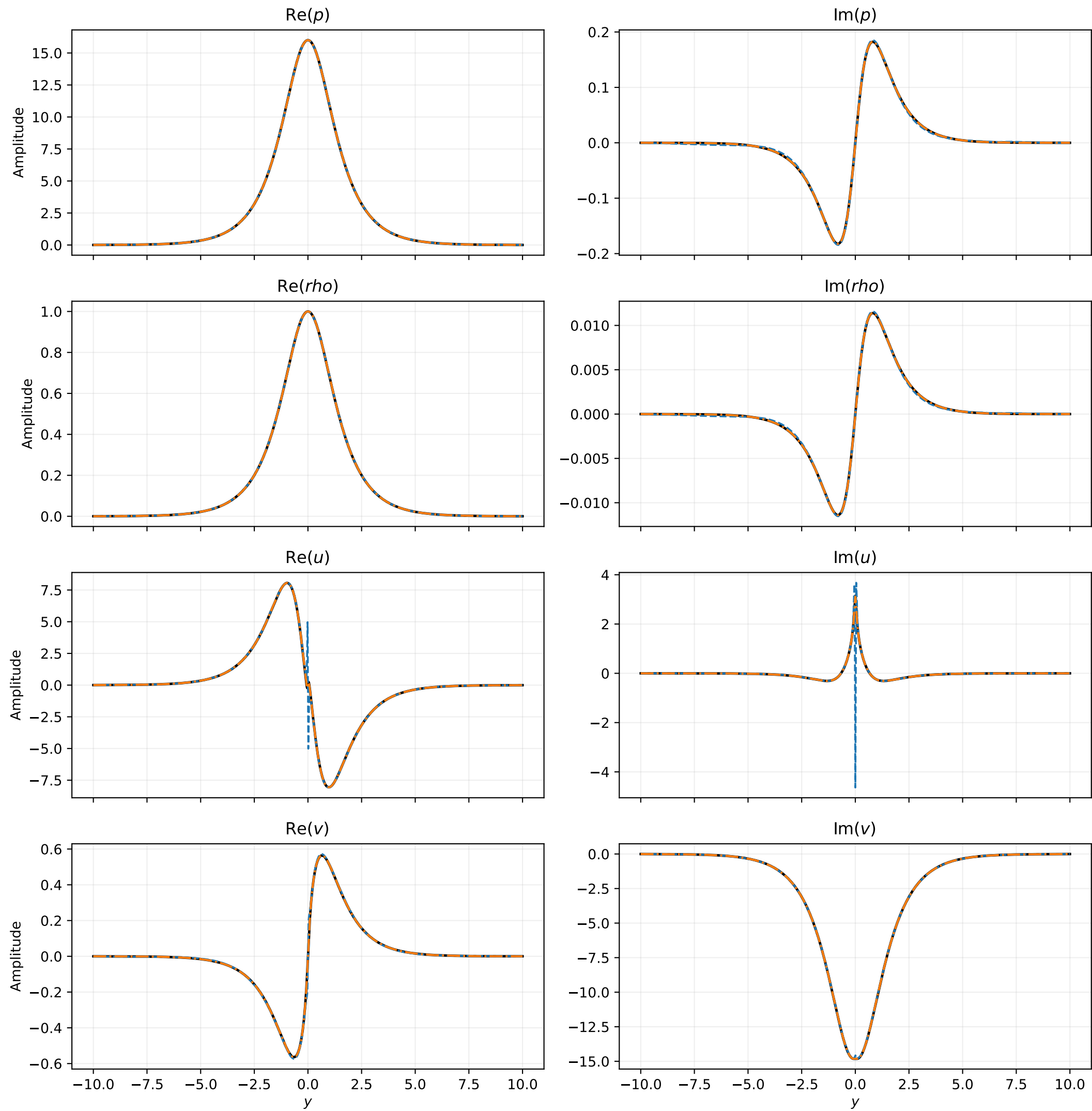
$$M = 0.10000, \eta = 0.32500, \alpha = 0.323371, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



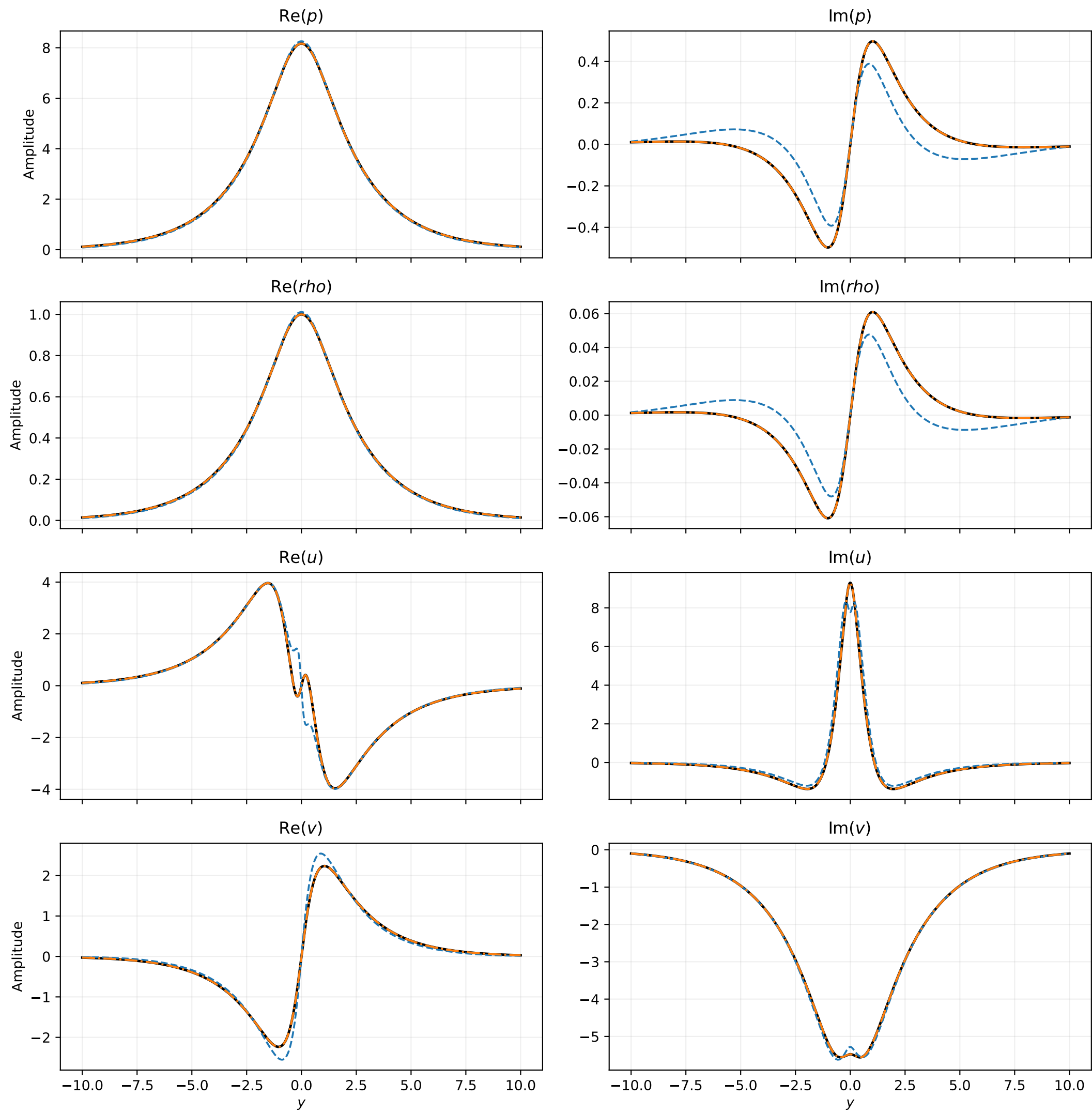
$M = 0.25000$, $\eta = 0.95000$, $\alpha = 0.919834$, $N = 501$

— Classical - - - Direct PINN - . - PINN + GEP



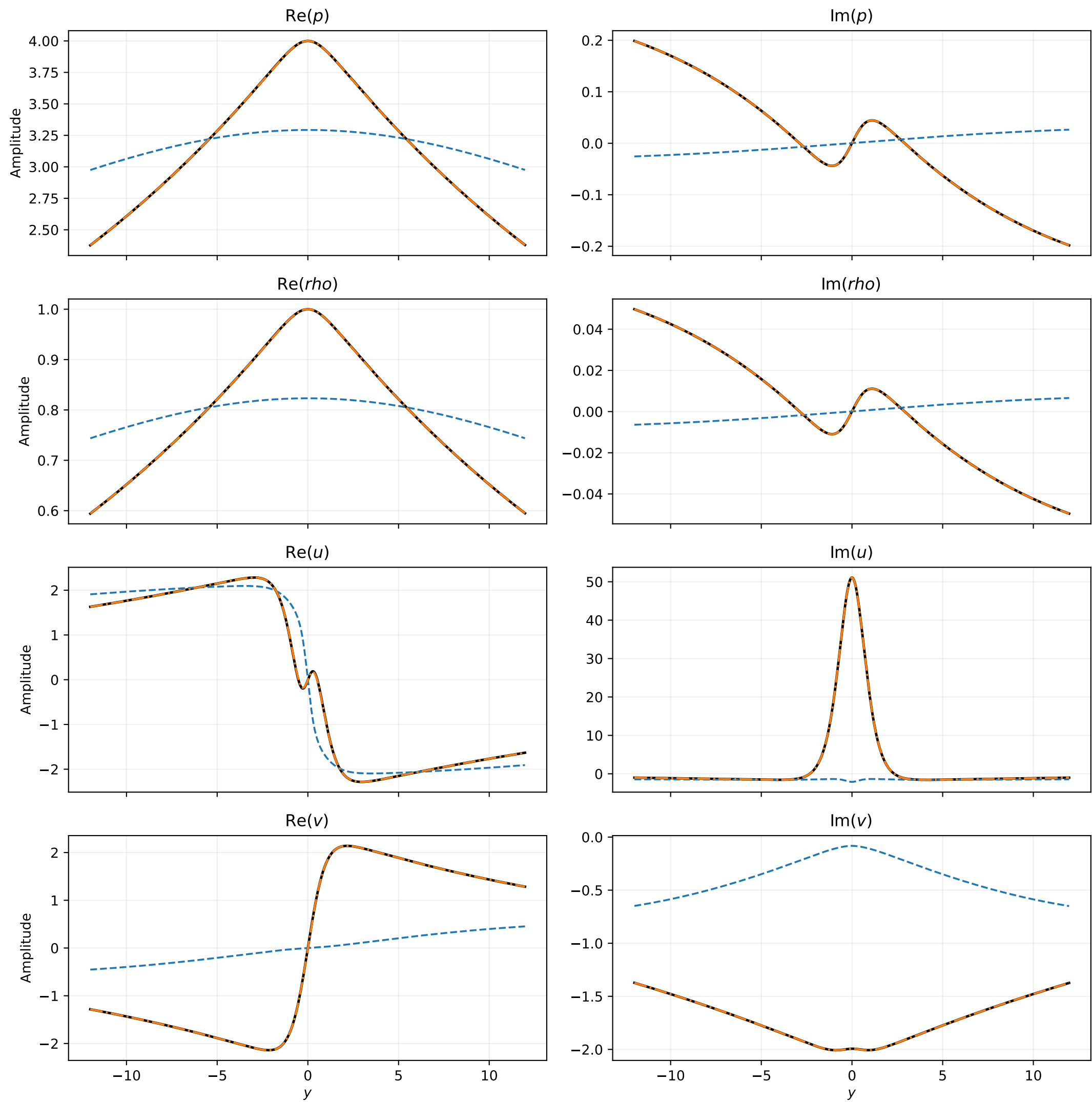
$$M = 0.35000, \eta = 0.51875, \alpha = 0.485939, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



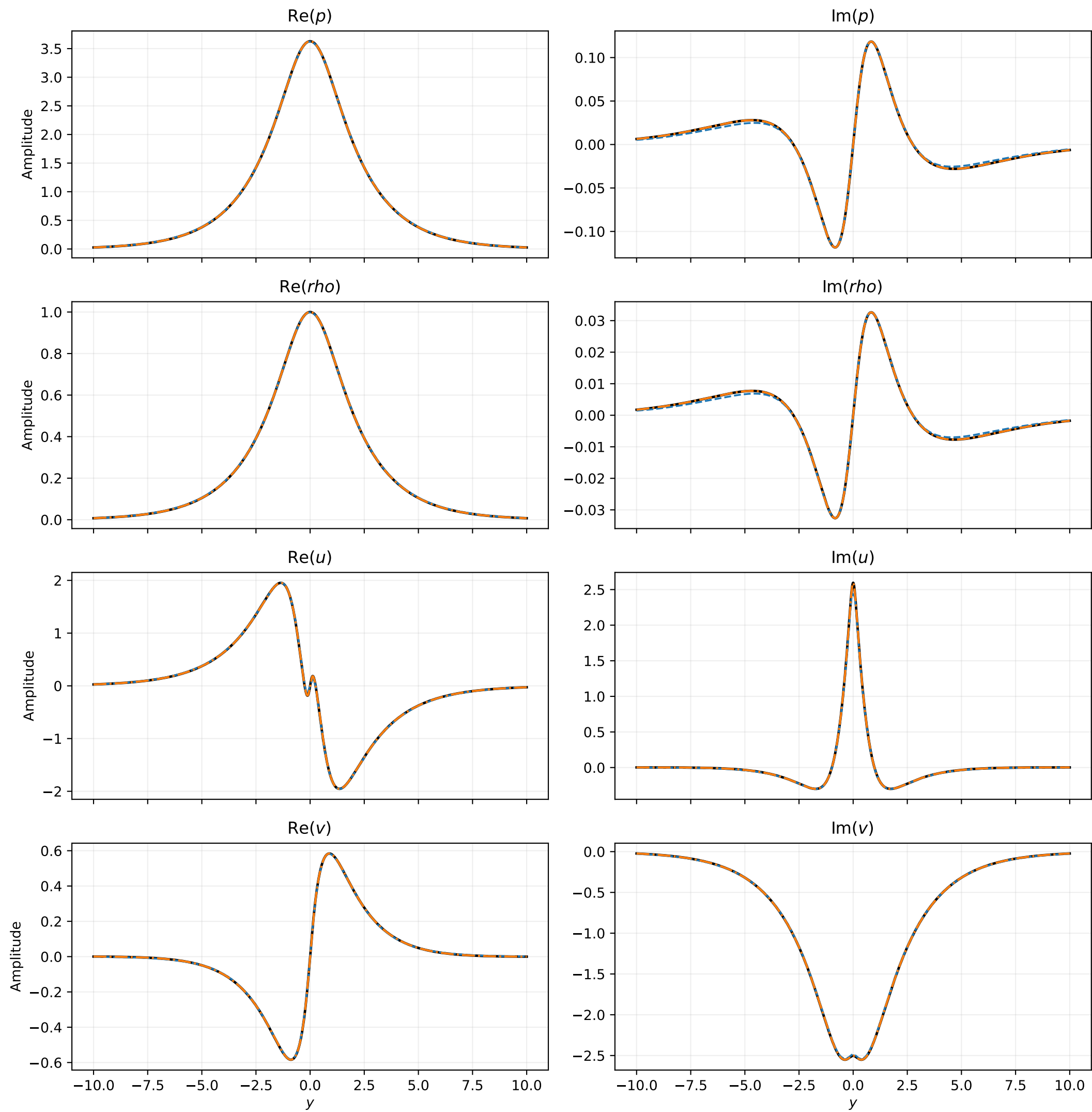
$M = 0.50000$, $\eta = 0.05500$, $\alpha = 0.047631$, $N = 401$

— Classical - - - Direct PINN - . - PINN + GEP



$$M = 0.52500, \eta = 0.72500, \alpha = 0.617049, N = 401$$

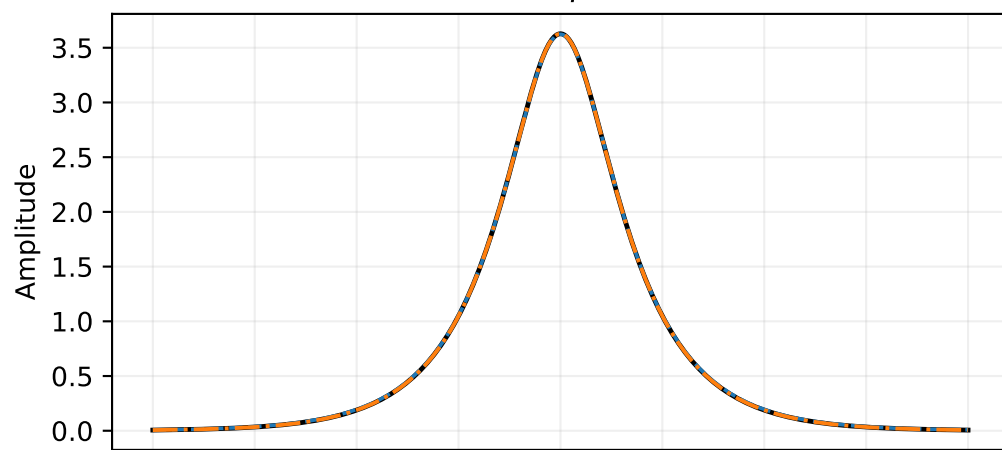
— Classical - - - Direct PINN - . - PINN + GEP



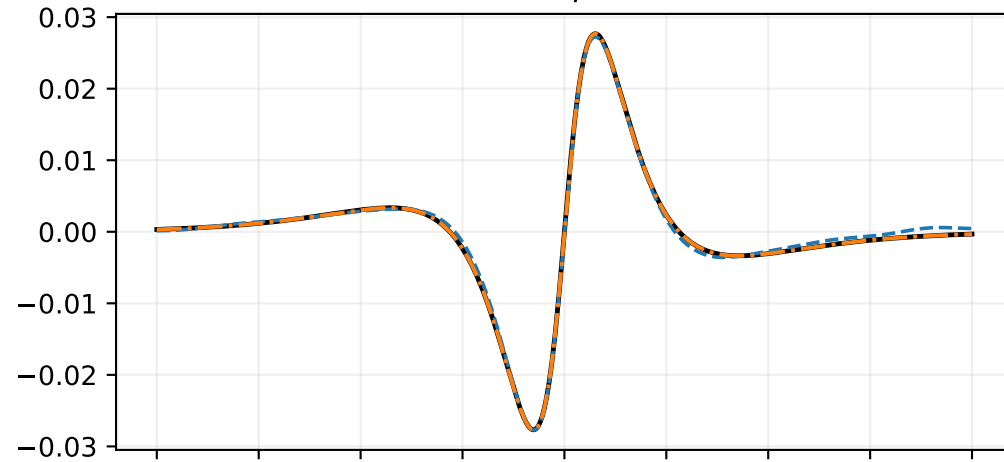
$$M = 0.52500, \eta = 0.95000, \alpha = 0.808547, N = 501$$

— Classical - - - Direct PINN - . - PINN + GEP

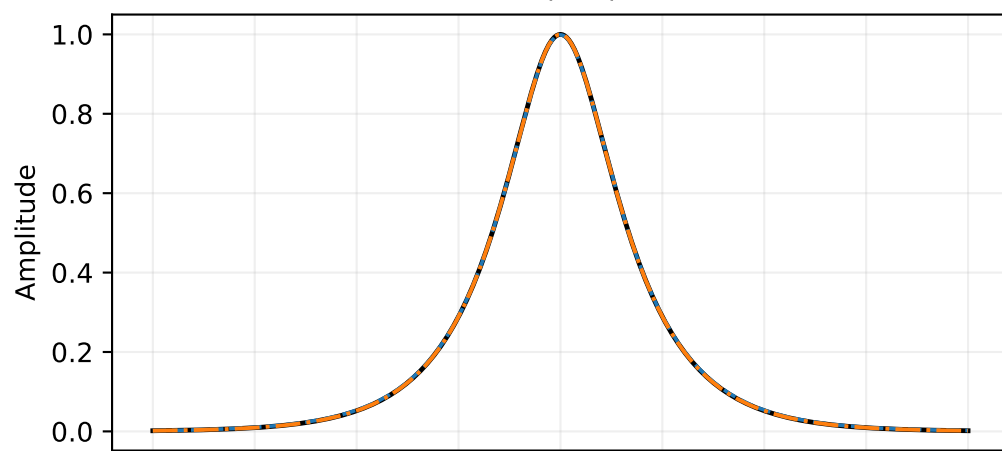
Re(p)



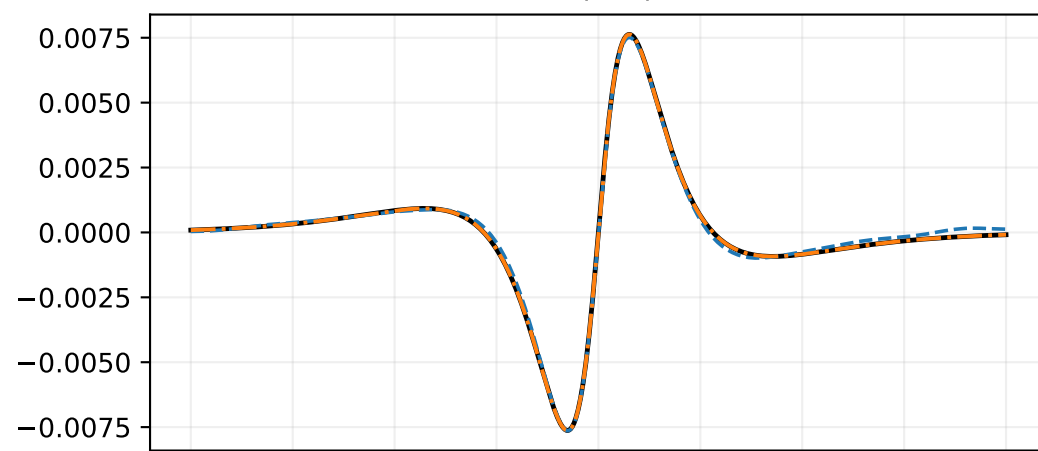
Im(p)



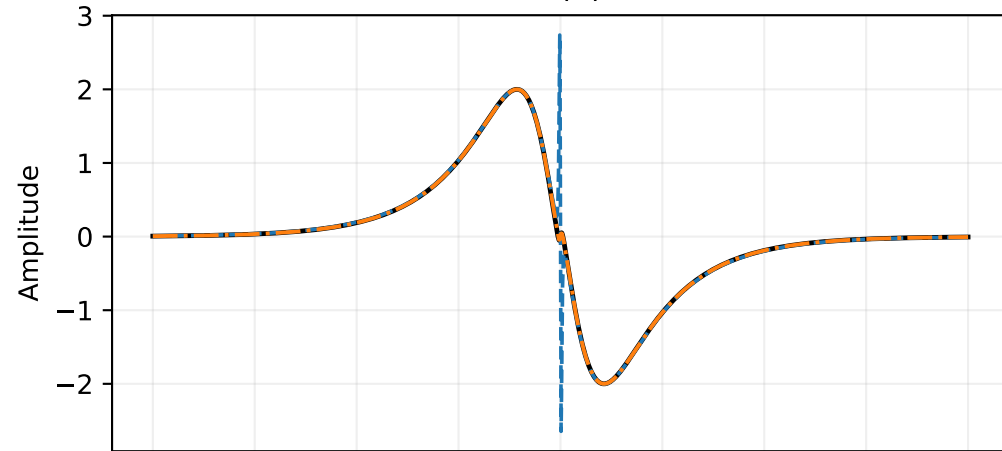
Re(ρ)



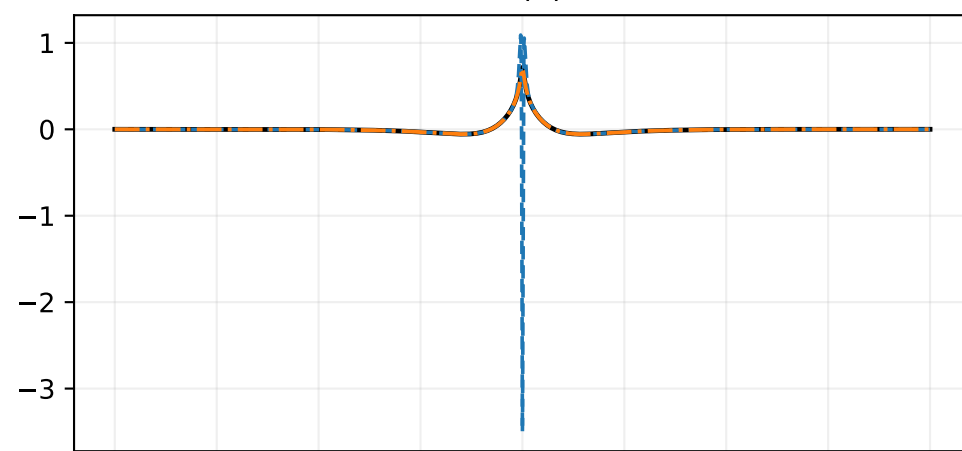
Im(ρ)



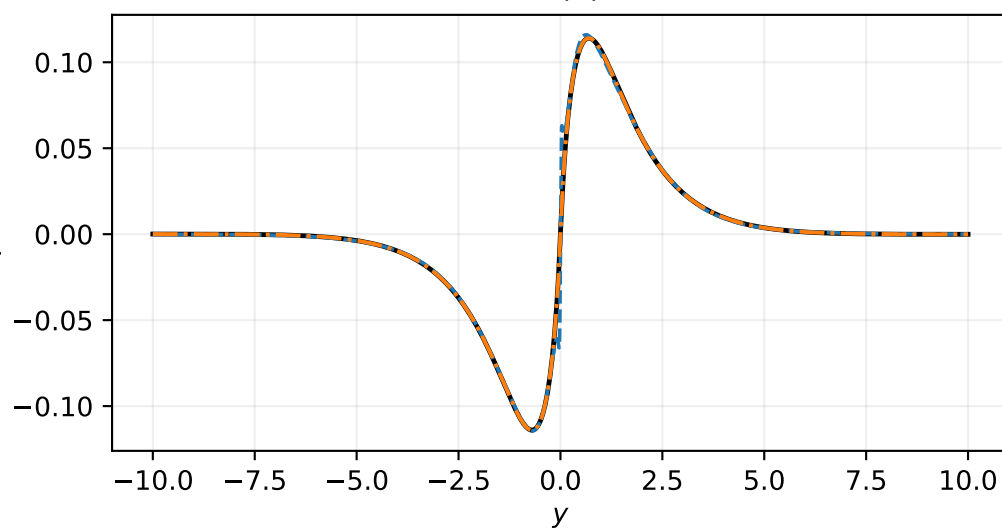
Re(u)



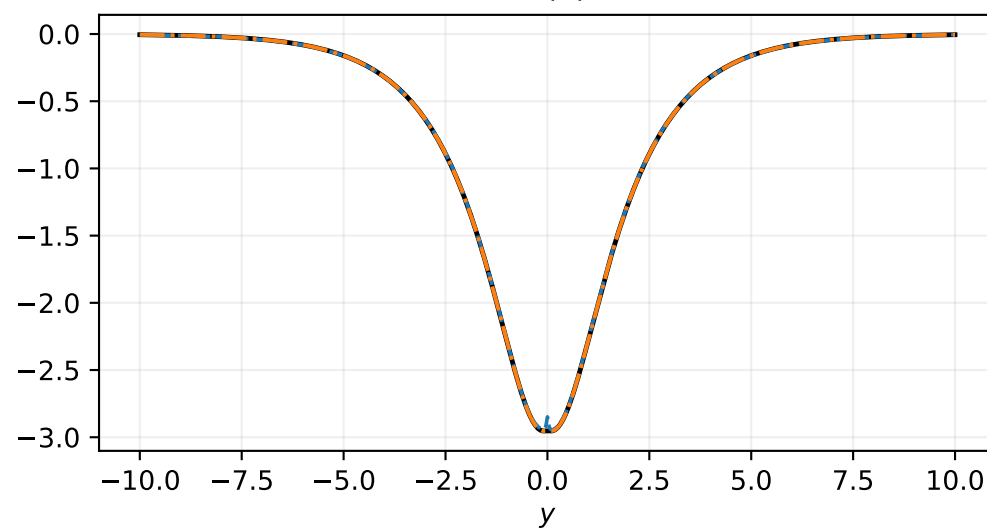
Im(u)



Re(v)

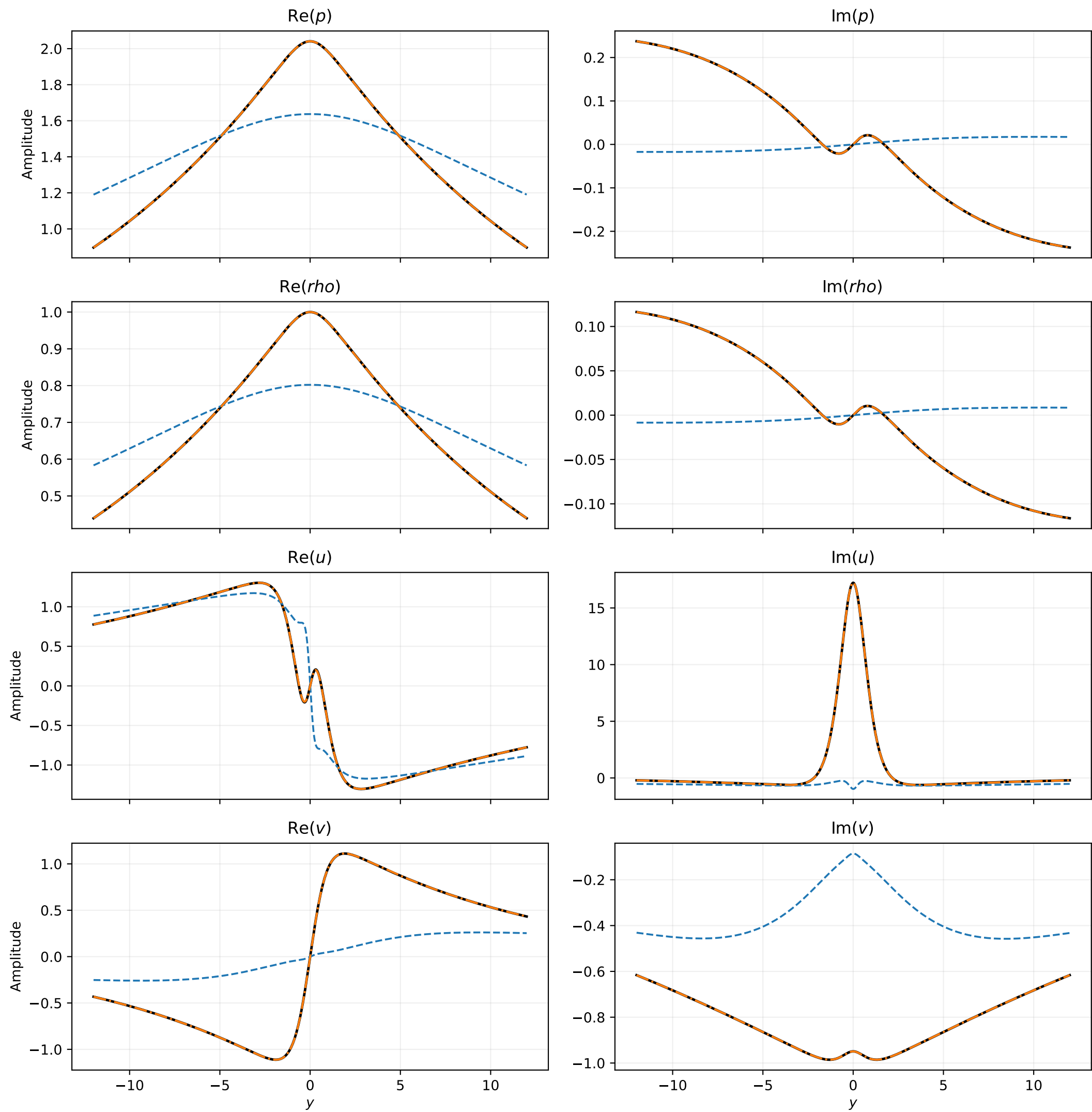


Im(v)



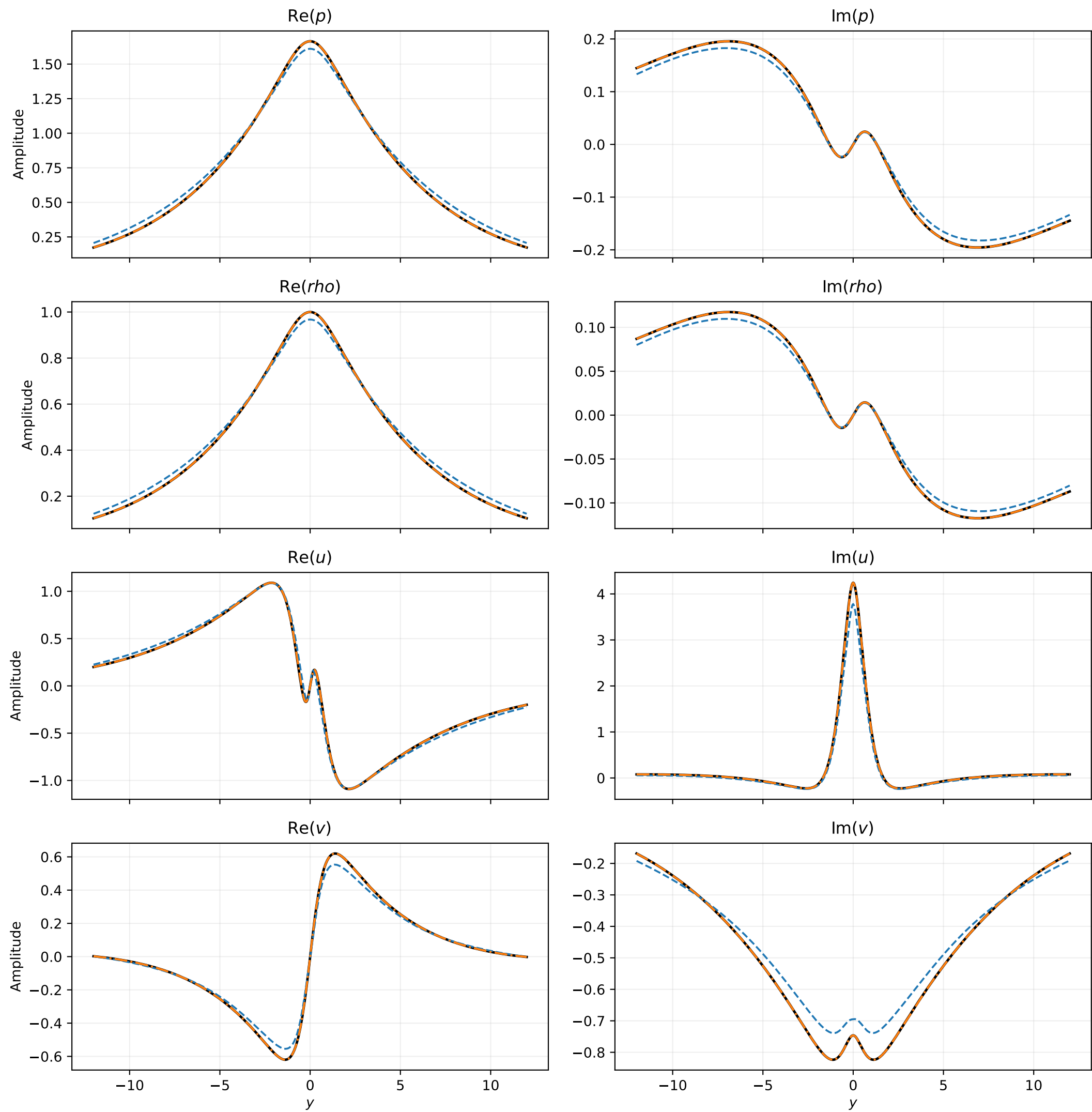
$$M = 0.70000, \eta = 0.11125, \alpha = 0.079448, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



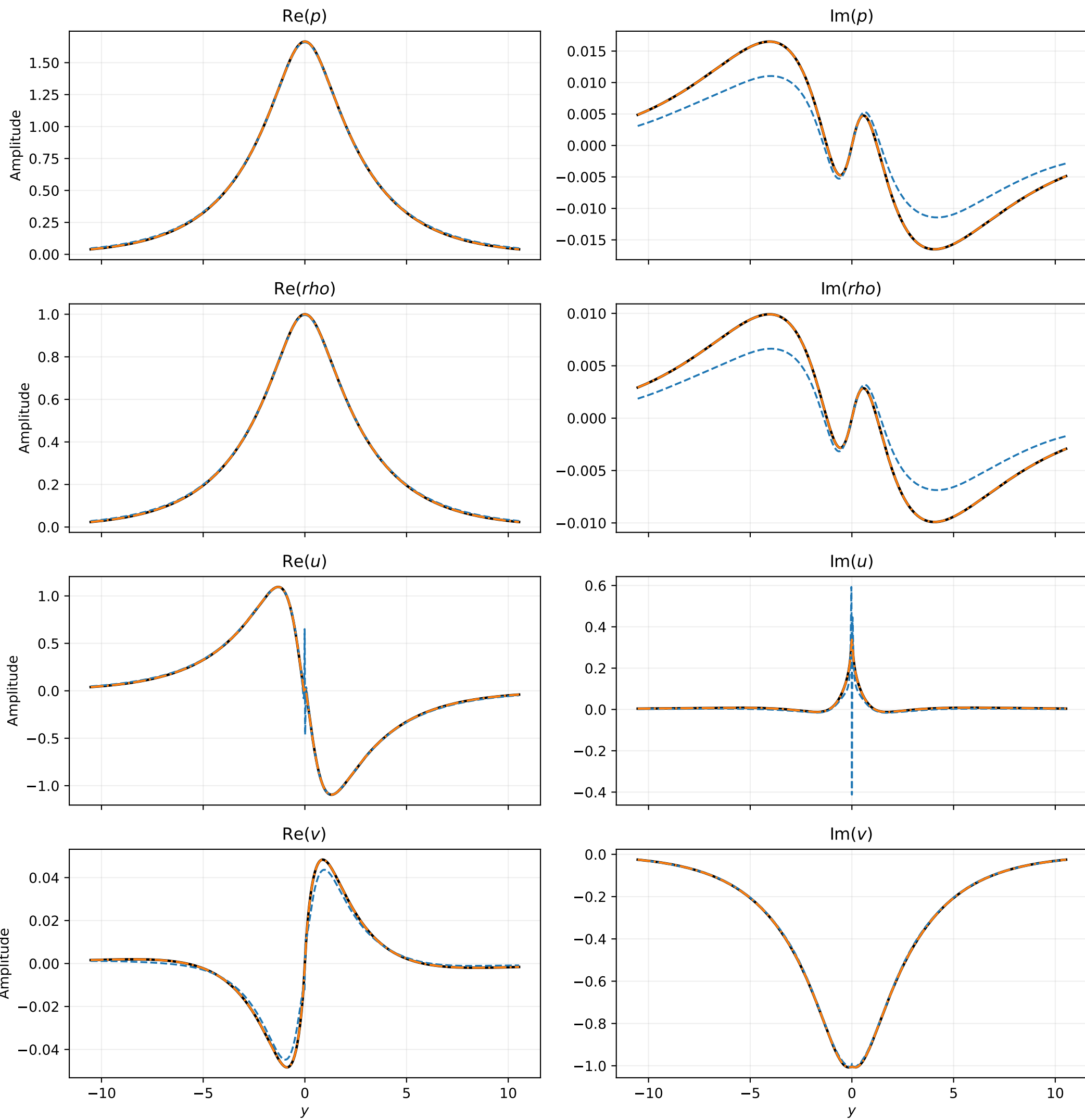
$$M = 0.77500, \eta = 0.38125, \alpha = 0.240935, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



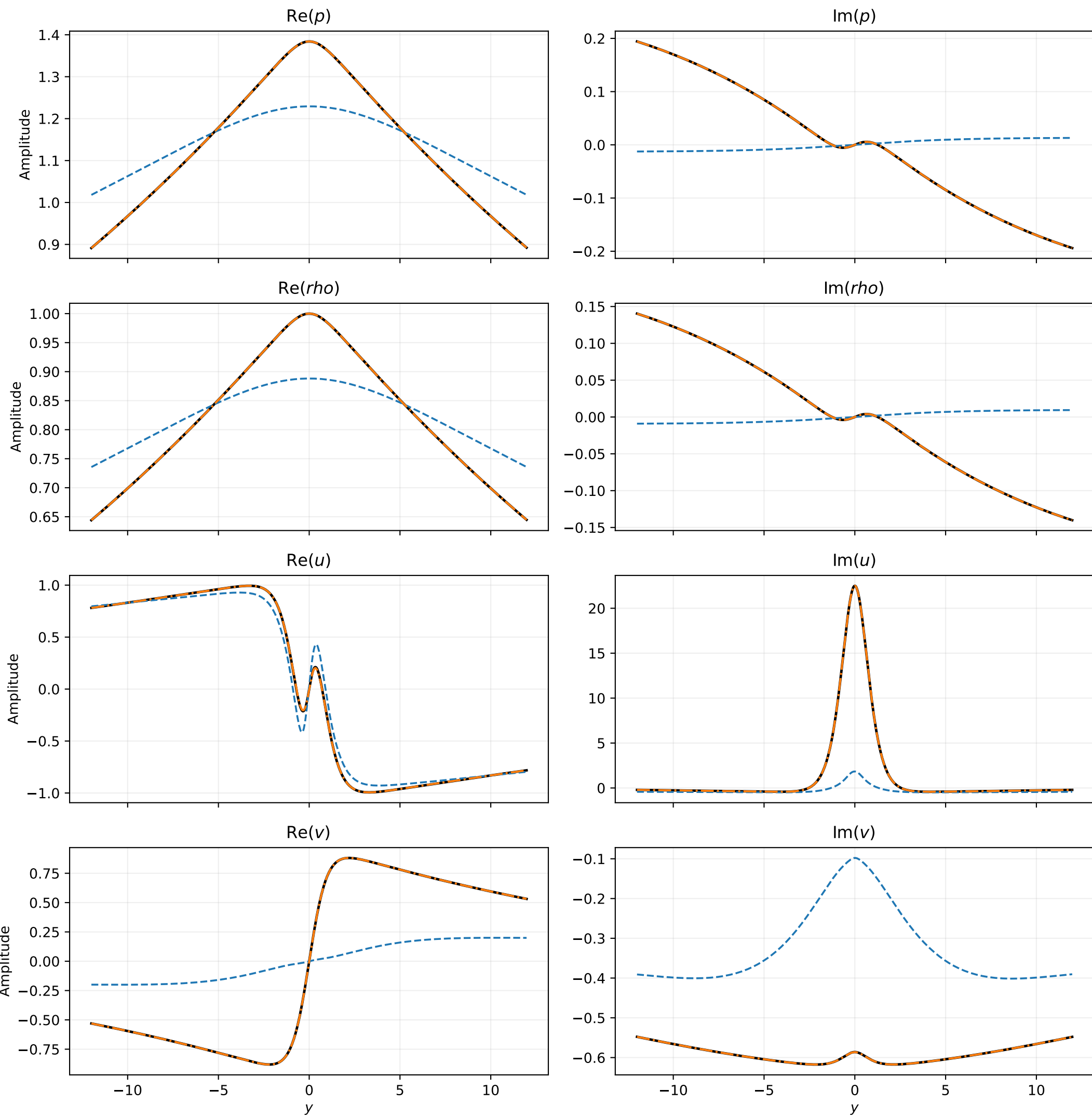
$$M = 0.77500, \eta = 0.95000, \alpha = 0.600363, N = 501$$

— Classical - - - Direct PINN - . - PINN + GEP



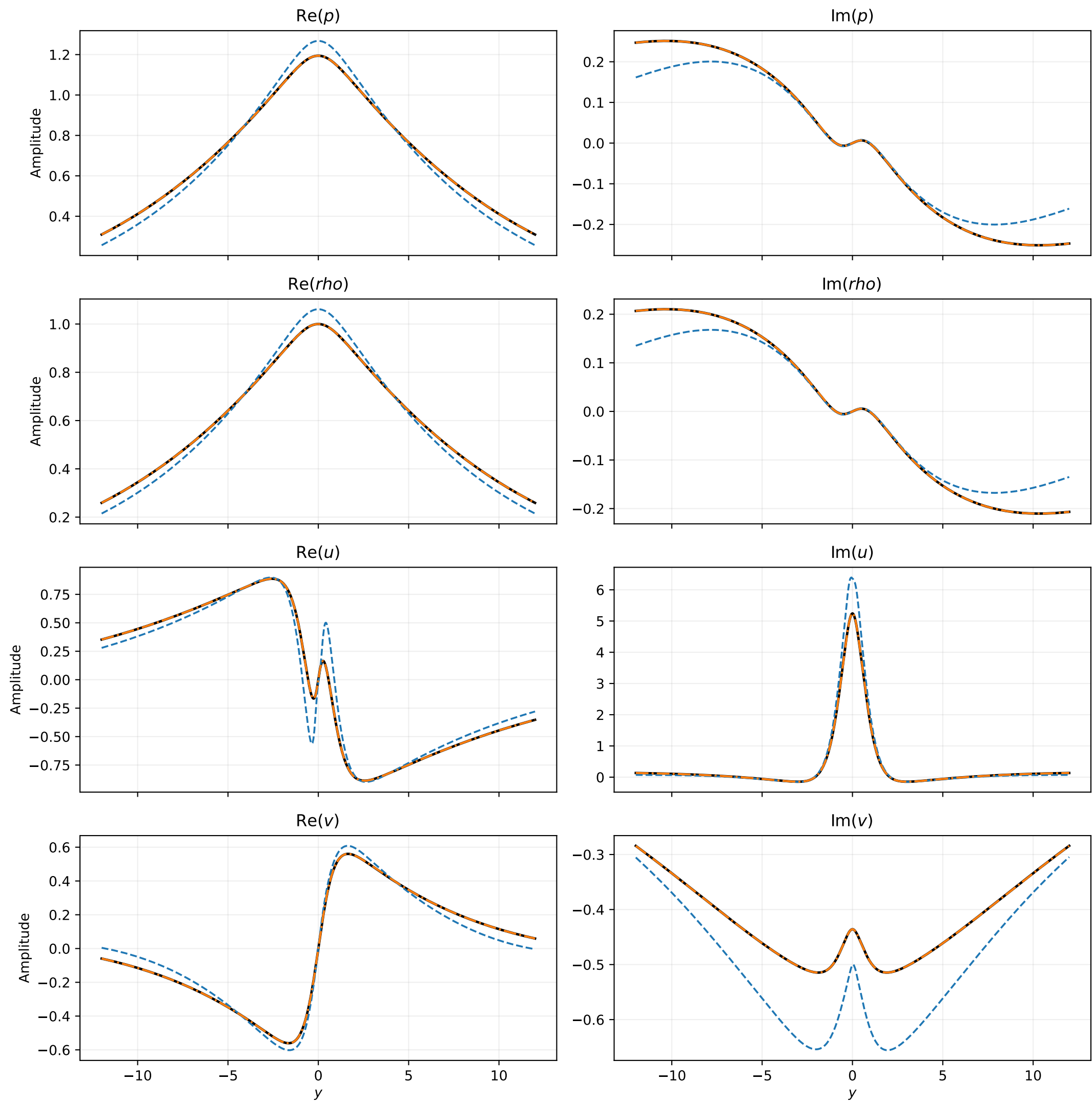
$$M = 0.85000, \eta = 0.08500, \alpha = 0.044777, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



$$M = 0.91500, \eta = 0.37500, \alpha = 0.151295, N = 401$$

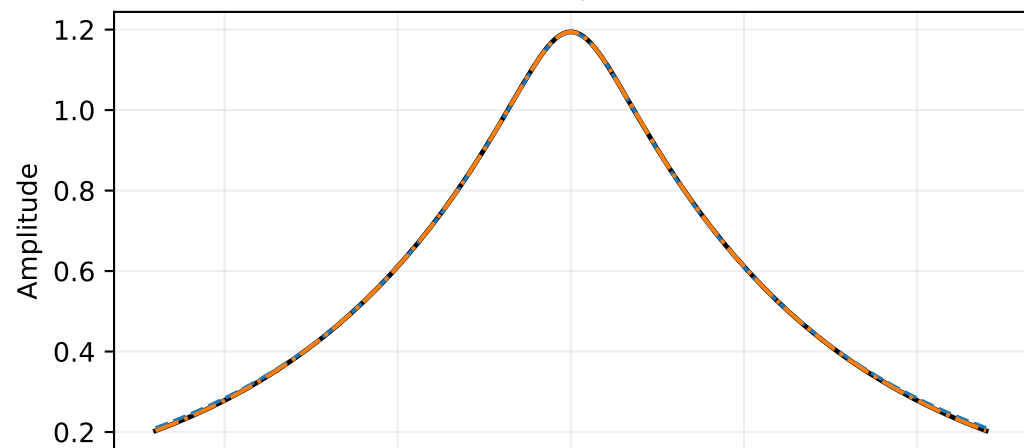
— Classical - - - Direct PINN - . - PINN + GEP



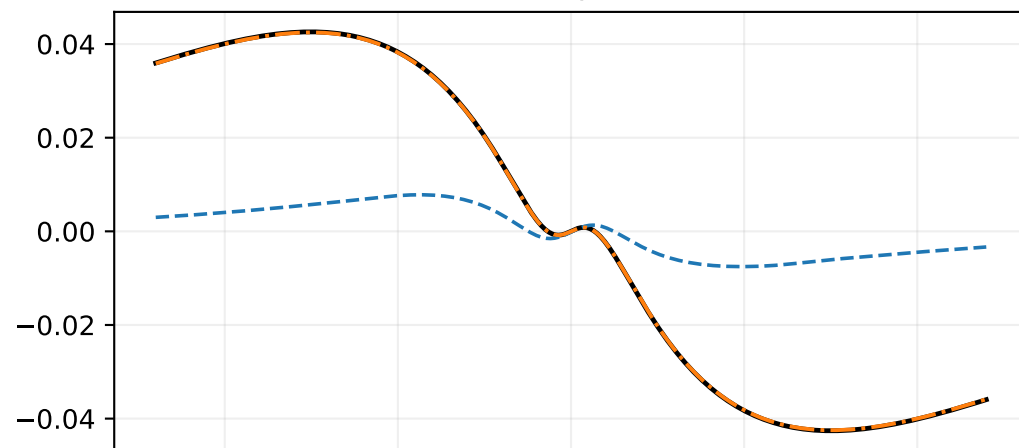
$$M = 0.91500, \eta = 0.95000, \alpha = 0.383281, N = 501$$

— Classical - - - Direct PINN - . - PINN + GEP

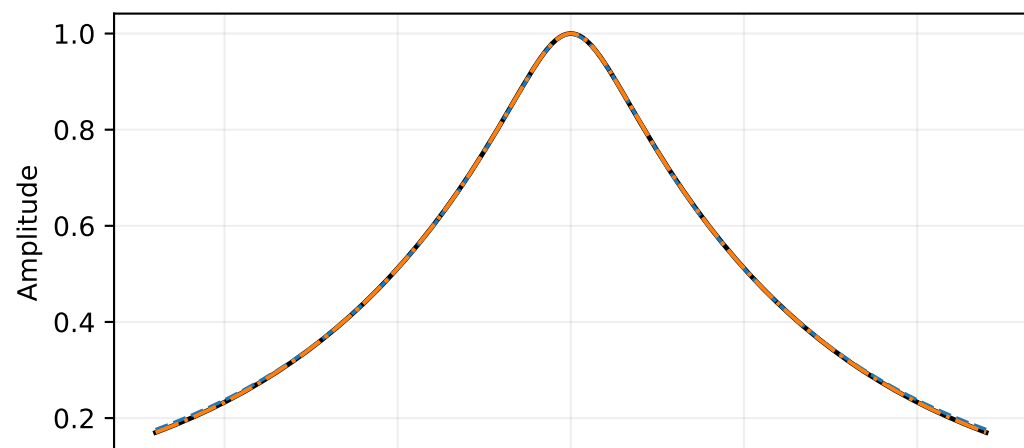
Re(p)



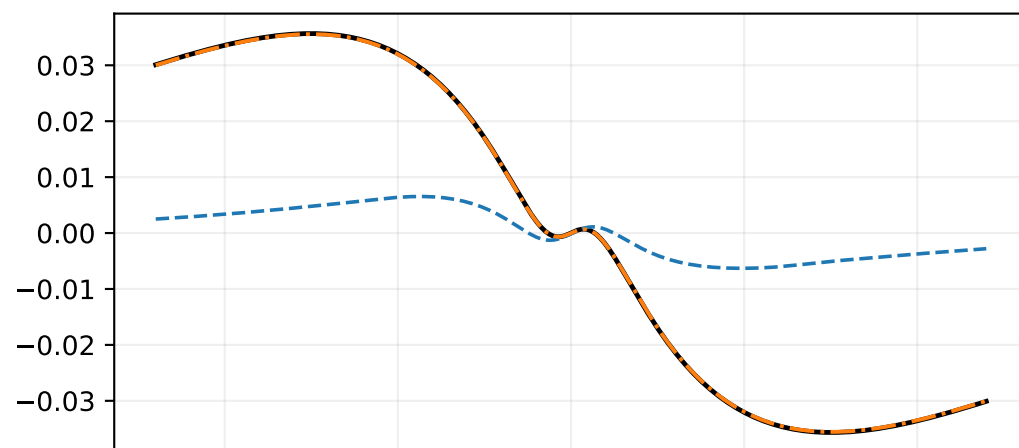
Im(p)



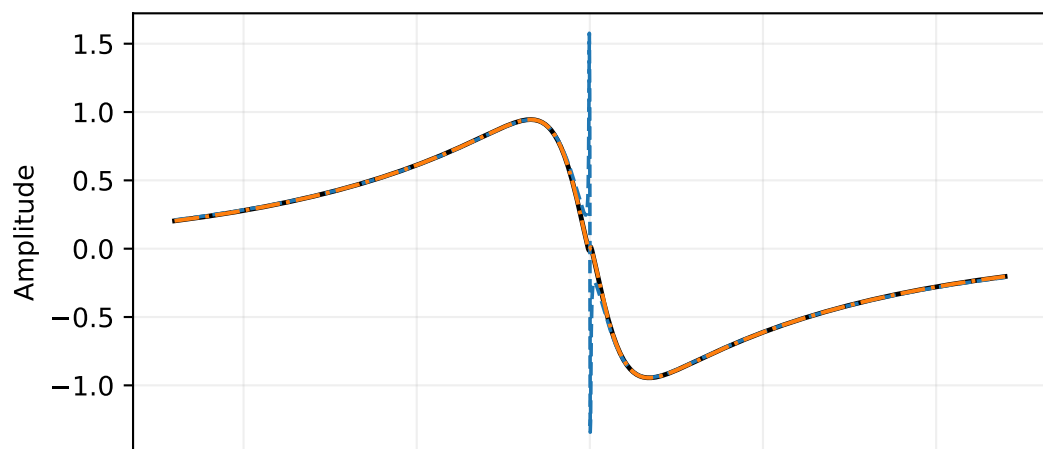
Re(ρ)



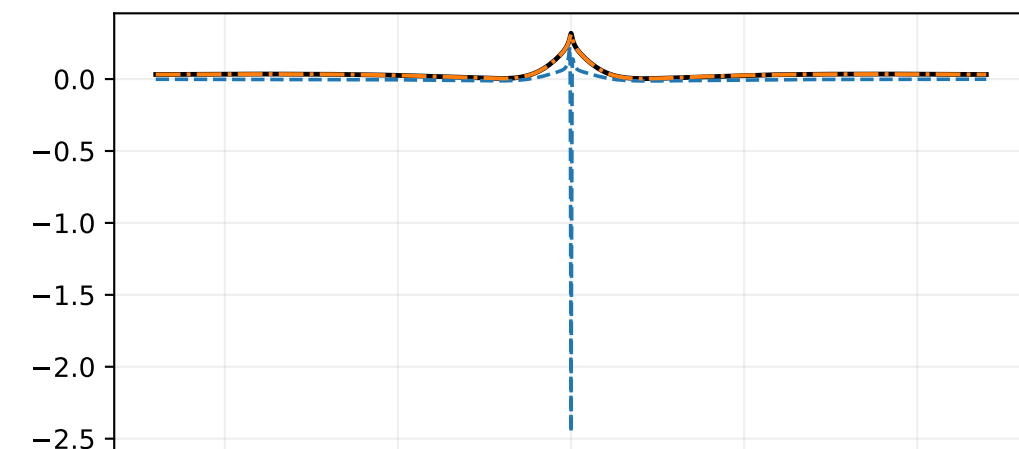
Im(ρ)



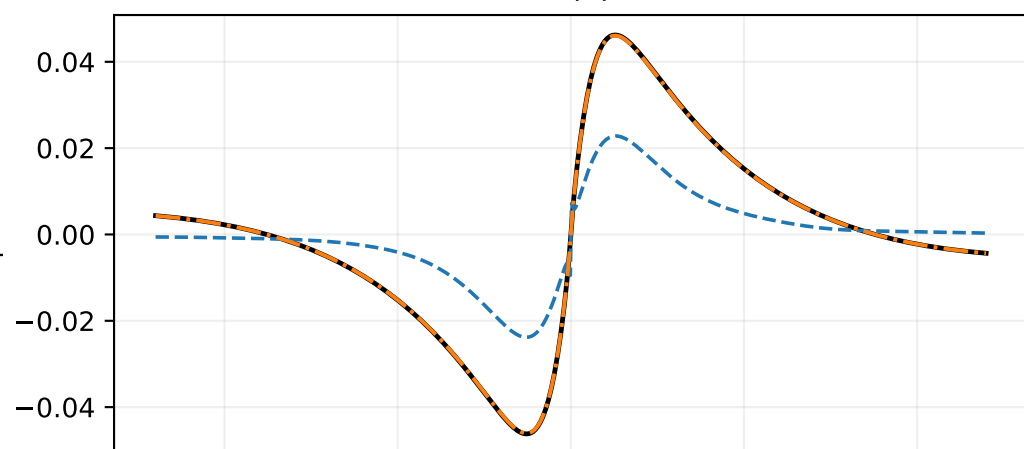
Re(u)



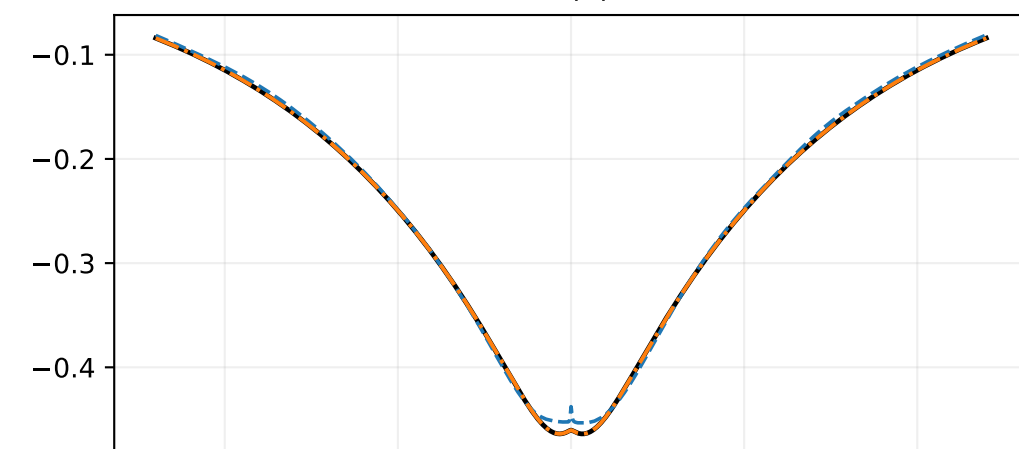
Im(u)



Re(v)

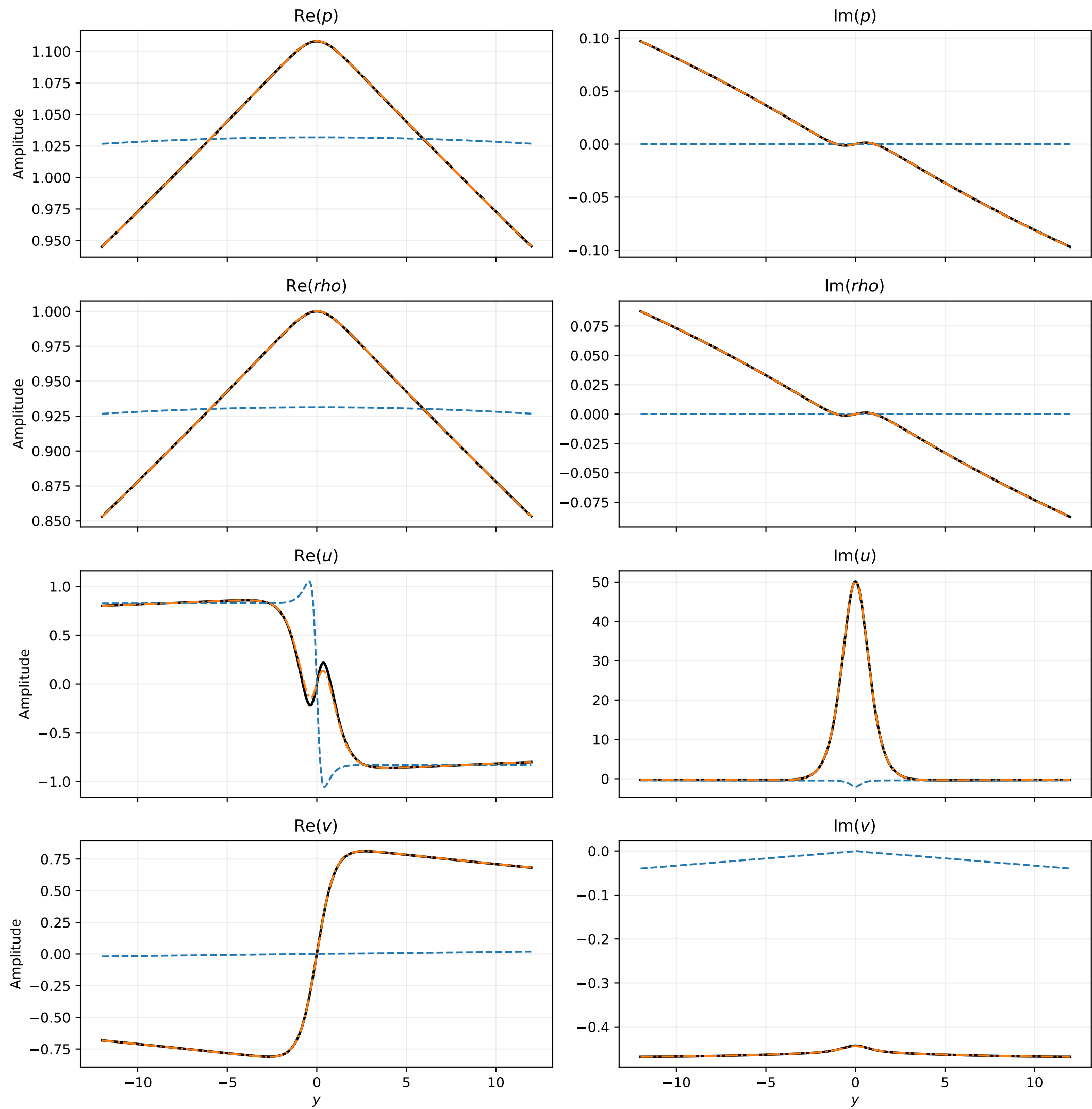


Im(v)



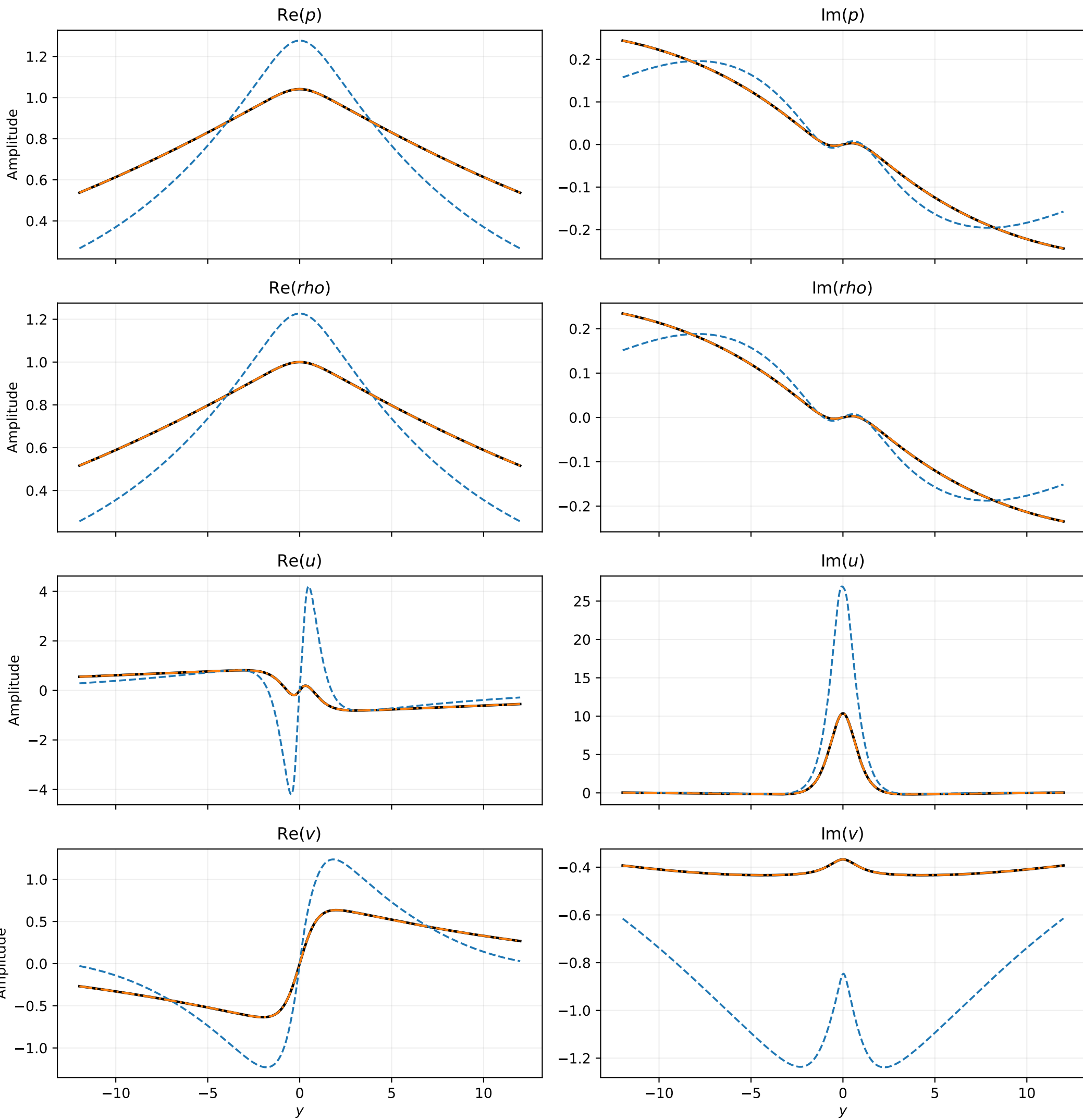
$$M = 0.95000, \eta = 0.05500, \alpha = 0.017174, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



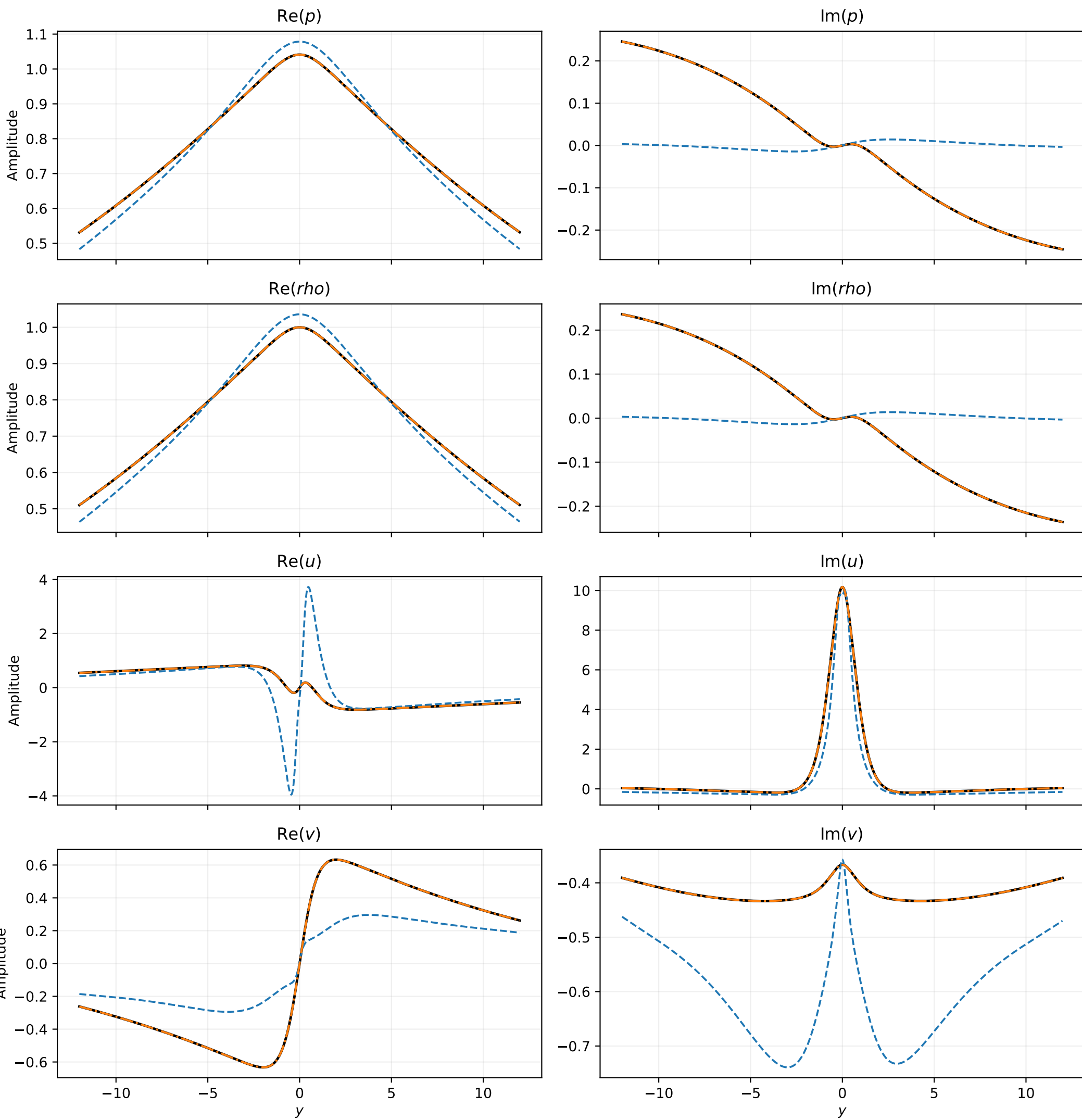
$$M = 0.98000, \eta = 0.37500, \alpha = 0.074624, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



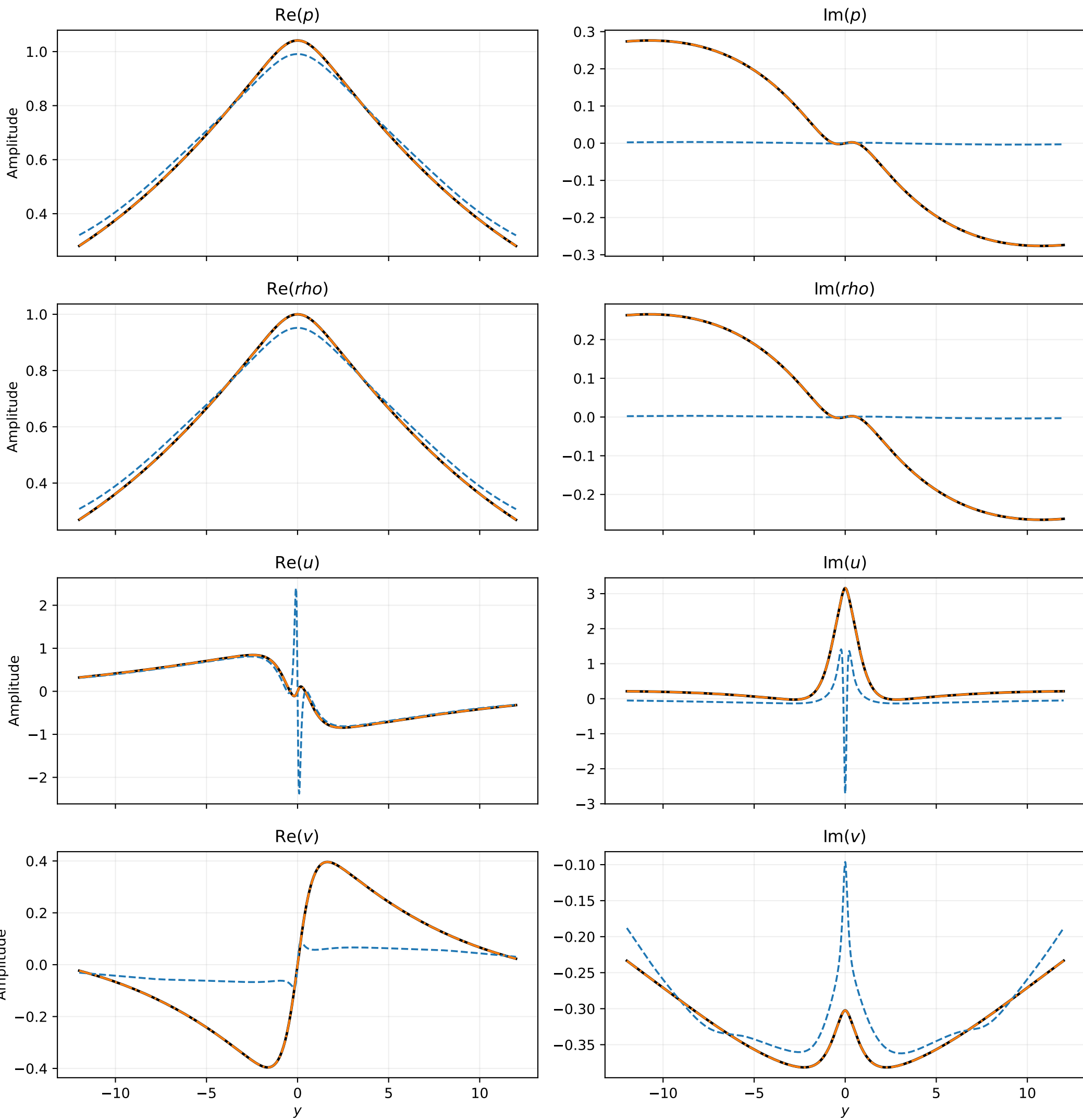
$$M = 0.98000, \eta = 0.38125, \alpha = 0.075868, N = 401$$

— Classical - - - Direct PINN - . - PINN + GEP



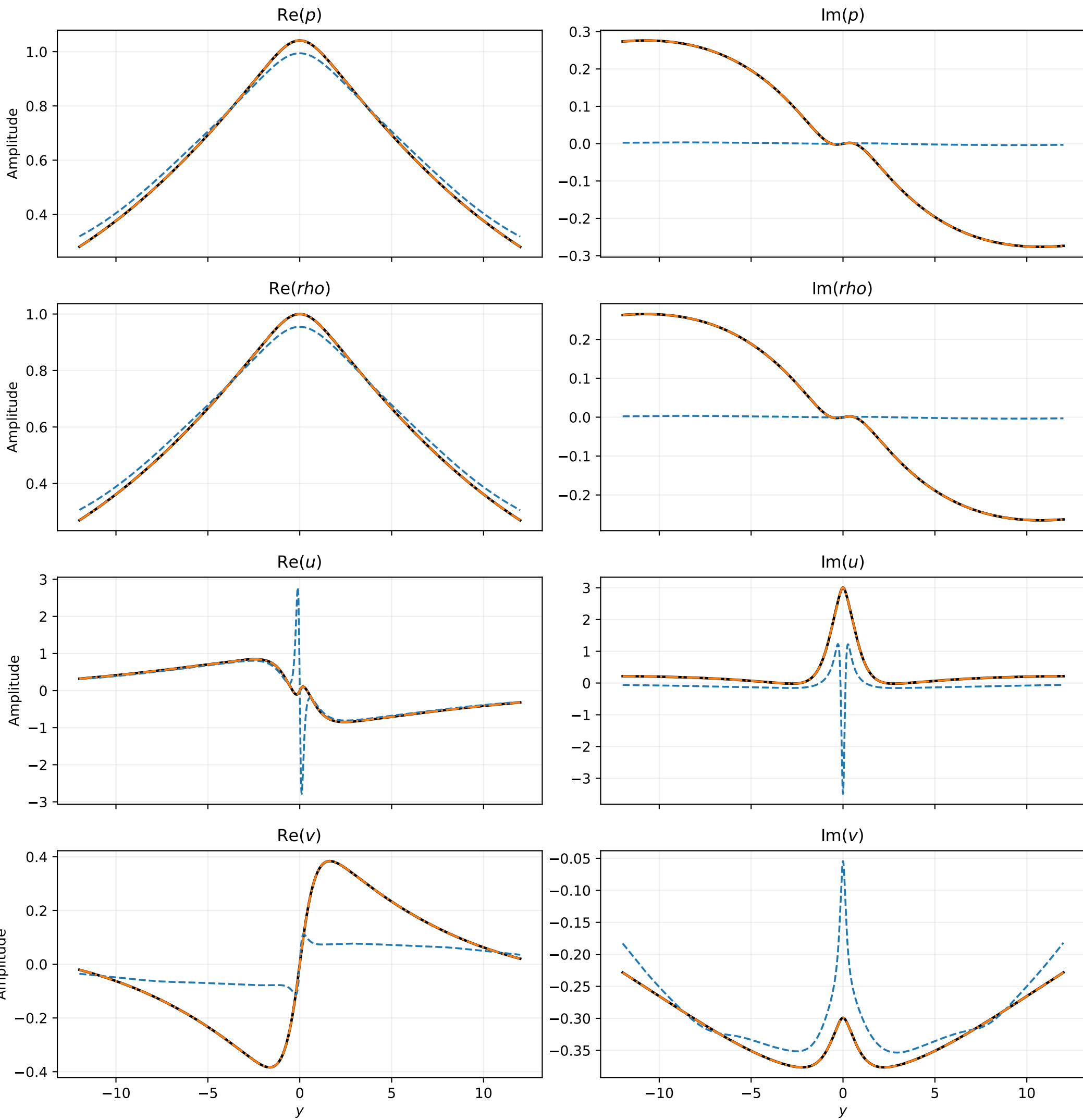
$$M = 0.98000, \eta = 0.92750, \alpha = 0.184570, N = 501$$

— Classical - - - Direct PINN - . - PINN + GEP



$$M = 0.98000, \eta = 0.95000, \alpha = 0.189048, N = 501$$

— Classical - - - Direct PINN - . - PINN + GEP



$$M = 0.98000, \eta = 0.97250, \alpha = 0.193525, N = 501$$

— Classical - - - Direct PINN - . - PINN + GEP

